SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

# CO2 - ASSESSMENT 1

## REPORT ANALYSIS

| | | |
|---|---|---|
| **Course Faculty** | **:** | **Dr. K.Anbazhagan** |
| **Course Name** | **:** | **Ethical Hacking** |
| **Course Code** | **:** | **ITA1406** |
| **Slot** | **:** | **Slot B** |
| **Total Marks** | **:** | **100** |

**CO2 :**

*Foot printing Tools: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT3)*

***Code of Conduct:***

*I **M.Chandana(192472250)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: Malli Chandana***

---

**Answer the Following**

Kindly follow the below step to do Domain & Email Footprinting using theHarvester

Step 1: Install Python

- Open https://www.python.org/downloads/
- Download the latest Python version.
- Run the installer.
- Tick Add Python to PATH
- Click Install Now

Step 2: Verify Python Installation

- Press Windows + R
- Type cmd and press Enter
- Run: [python -m pip install --upgrade pip] in cmd

Step 4: Install theHarvester

- Run: [pip install theHarvester] in cmd
- Run: [python -m pip install theHarvester] in cmd

- Wait until the installation Tinishes.

Step 5: Verify theHarvester

- Run: [theHarvester --help] in cmd

Step 6: Run theHarvester

- Run: [theHarvester -d saveetha.com -b all] in cmd

Step 7: Wait for the Results

- After a few minutes, theHarvester displays information such as:
- Email Addresses
- Subdomains
- Host Names
- IP Addresses

**Fill the following required details**

| | |
|---|---|
| Domain Name | |
| Emails found (List all one by one) | |
| Sub emails found (List all one by one) | |
| Hosts and Its IP (List all one by one) | |
| Total Email Addresses Found | |
| Total Subdomains Found | |
| Total Hosts Found | |
| Total IP Addresses Found | |
| Screenshot of Commend execution | |
| Screenshot of theHarvester output | |

# AIM AND INTRODUCTION

## 1. Aim

To perform domain and email footprinting on the domain `saveetha.com` using theHarvester tool and collect publicly available information such as email addresses, IP addresses, hostnames, URLs, and Autonomous System Numbers (ASNs).

## 2. Introduction

Footprinting is the first stage of ethical hacking. It is the process of collecting information about a target organization, website, or computer network before conducting further security analysis.

Footprinting helps ethical hackers understand the publicly visible information about a target. This information may include domain names, IP addresses, email addresses, subdomains, employee details, and publicly accessible web pages.

There are two major types of footprinting:

**1. Passive Footprinting:** Information is collected from publicly available sources without directly interacting with the target's systems in an intrusive way. Examples include search engines, public DNS records, and online databases.

**2. Active Footprinting:** Information is collected by directly interacting with the target system or network. Examples may include authorized network scanning and service identification.

In this experiment, theHarvester tool is used to gather publicly available information related to the domain `saveetha.com`.

## 3. About theHarvester

theHarvester is an open-source information-gathering tool used in cybersecurity and ethical hacking. It helps identify information associated with a domain by searching supported public sources.

The tool can collect information such as:

- Email addresses associated with a domain.
- IP addresses.
- Hostnames and subdomains.
- URLs.
- Autonomous System Numbers (ASNs).

TheHarvester is useful during the reconnaissance phase of a security assessment. The collected information can help security professionals understand an organization's public-facing digital footprint.

## 4. Requirements

**Hardware requirements:**

- Computer or laptop.
- Internet connection.

**Software requirements:**

- Windows operating system.
- Python.
- Git.
- `uv` package and project manager.
- theHarvester tool.
- Command Prompt or terminal.

# THEORY AND WORKING PRINCIPLE

## 5. Theory

Domain footprinting is the process of gathering information related to a domain name. A domain name is a human-readable address used to access a website, such as `saveetha.com`.

Email footprinting focuses on identifying email addresses associated with a target domain through public information sources. These addresses may help an organization understand its exposure to phishing and other email-based threats.

IP addresses identify network interfaces or hosts on a network. Hostnames are names associated with systems or services. URLs identify web resources, while ASNs identify autonomous systems participating in Internet routing.

TheHarvester automates the collection of several types of information by querying the public sources it supports. The amount of information returned depends on source availability, search limitations, API credentials, and network connectivity.

## 6. Information Collected

| Information | Description |
|---|---|
| Domain name | The domain selected for investigation |
| Email addresses | Publicly discovered email addresses related to the domain |
| IP addresses | IP addresses identified by the tool |
| Hostnames | Hostnames and subdomains associated with the target |
| URLs | Web addresses discovered during collection |
| ASNs | Autonomous System Numbers associated with discovered network information |

## 7. Working Principle

The working process of theHarvester is as follows:

1. The user specifies the target domain.
2. The user selects the information sources to search.
3. theHarvester queries the supported sources.
4. The tool collects publicly available information.
5. The results are displayed in the terminal.
6. The collected information is reviewed and documented for security analysis.

**Command used:**

```
uv run theHarvester -d saveetha.com -b all
```

**Explanation of the command:**

- `uv run` executes the tool in the project's configured Python environment.
- `theHarvester` specifies the tool to execute.
- `-d saveetha.com` specifies the target domain.
- `-b all` requests all supported data sources.

The command may not retrieve information from every source. Some sources require API keys, and others may be unavailable or return errors.

# PROCEDURE

## 8. Step-by-Step Procedure

**Step 1: Install the required software**

Python, Git, and `uv` were used to prepare the environment required to run theHarvester.

**Step 2: Download theHarvester**

The official GitHub repository was cloned using the following command:

```
git clone https://github.com/laramies/theHarvester.git
```

This downloaded the project files to the computer.

**Step 3: Open the project folder**

The terminal was opened in the downloaded `theHarvester` project directory.

**Step 4: Install project dependencies**

The following command was executed:

```
uv sync
```

This installed the project dependencies and prepared the virtual environment.

**Step 5: Execute the footprinting command**

The following command was run in the terminal:

```
uv run theHarvester -d saveetha.com -b all
```

**Step 6: Collect the results**

The tool searched its supported information sources and displayed the discovered information, including ASNs, URLs, IP addresses, email addresses, and hostnames.

## Analyze the output

```
C:\Users\malli\theHarvester>uv run theHarvester -d saveetha.com -b all
*******************************************************************
*  _   _                                             _            *
* | |_| |__   ___    /\  /\__ _ _ ____   _____  ___| |_ ___ _ __  *
* | __|  _ \ / _ \  / /_/ / _` | '__\ \ / / _ \/ __| __/ _ \ '__| *
* | |_| | | |  __/ / __  / (_| | |   \ V /  __/\__ \ ||  __/ |    *
*  \__|_| |_|\___| \/ /_/ \__,_|_|    \_/ \___||___/\__\___|_|    *
*                                                                 *
* theHarvester 5.0.0                                              *
* Coded by Christian Martorella                                   *
* Edge-Security Research                                          *
* cmartorella@edge-security.com                                   *
*                                                                 *
*******************************************************************
[*] Activity: P0 passive collection

[*] Target: saveetha.com

[*] Source workers: requested=3; effective=3.
[*] Searching Apis-guru.
[*] Searching Arquivo.
[*] Searching Baidu.
ERROR asyncio: Task exception was never retrieved
future: <Task finished name='Task-9' coro=<Connection.run() done, defined at C:\Users\malli\theHarvester\.venv\Lib\site-packages\play
wright\_impl\_connection.py:305> exception=ValueError('startupinfo is not supported')>
Traceback (most recent call last):
  File "C:\Users\malli\theHarvester\.venv\Lib\site-packages\playwright\_impl\_connection.py", line 312, in run
    await self._transport.connect()
  File "C:\Users\malli\theHarvester\.venv\Lib\site-packages\playwright\_impl\_transport.py", line 133, in connect
    raise exc
  File "C:\Users\malli\theHarvester\.venv\Lib\site-packages\playwright\_impl\_transport.py", line 120, in connect
    self._proc = await asyncio.create_subprocess_exec(
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    ...<9 lines>...
```

```
Command Prompt
[*] Searching Thc.
[!] Source tomba skipped: missing credentials.
[*] Searching Urlscan.
[!] Source virustotal skipped: missing credentials.
[*] Searching Waybackarchive.
WARNING theHarvester.discovery.commoncrawl: Common Crawl API error for index CC-MAIN-2026-34: Expecting value: line 1 column 1 (char
0)
[!] Source whoisxml skipped: missing credentials.
[*] Searching Windvane.
[*] Searching Yahoo.
WARNING theHarvester.discovery.commoncrawl: Common Crawl API error for index CC-MAIN-2026-34: Expecting value: line 1 column 1 (char
0)
[!] Source zoomeye skipped: missing credentials.
WARNING theHarvester.discovery.commoncrawl: Common Crawl API error for index CC-MAIN-2026-25: Expecting value: line 1 column 1 (char
0)
WARNING theHarvester.discovery.commoncrawl: Common Crawl API error for index CC-MAIN-2026-21: Expecting value: line 1 column 1 (char
0)

[*] ASNS found: 2
--------------------
AS16509
AS53831

[*] ASN organization attributions found: 3
-----------------------------------
AS16509 | AMAZON-02 - Amazon.com, Inc., US | source:urlscan | ip:3.110.122.44
AS53831 | SQUARESPACE - Squarespace, Inc., US | source:urlscan | hostname:www.saveetha.com
AS53831 | SQUARESPACE - Squarespace, Inc., US | source:urlscan | ip:198.185.159.144

[*] URLs found: 2
--------------------
http://saveetha.com/
https://www.saveetha.com/

[*] IPs found: 23
-------------------
103.249.82.245
```

```
AS53831 | SQUARESPACE - Squarespace, Inc., US | source:urlscan | ip:198.185.159.144

[*] URLs found: 2
--------------------
http://saveetha.com/
https://www.saveetha.com/

[*] IPs found: 23
-------------------
103.249.82.245
103.249.82.249
106.51.116.185
118.139.175.1
14.139.187.227
14.98.51.2
180.235.121.243
182.19.38.181
198.185.159.144
198.185.159.145
198.49.23.144
198.49.23.145
199.34.228.77
3.110.122.44
34.149.87.45
43.204.251.6
43.205.1.154

[*] No emails found.

[*] No people found.

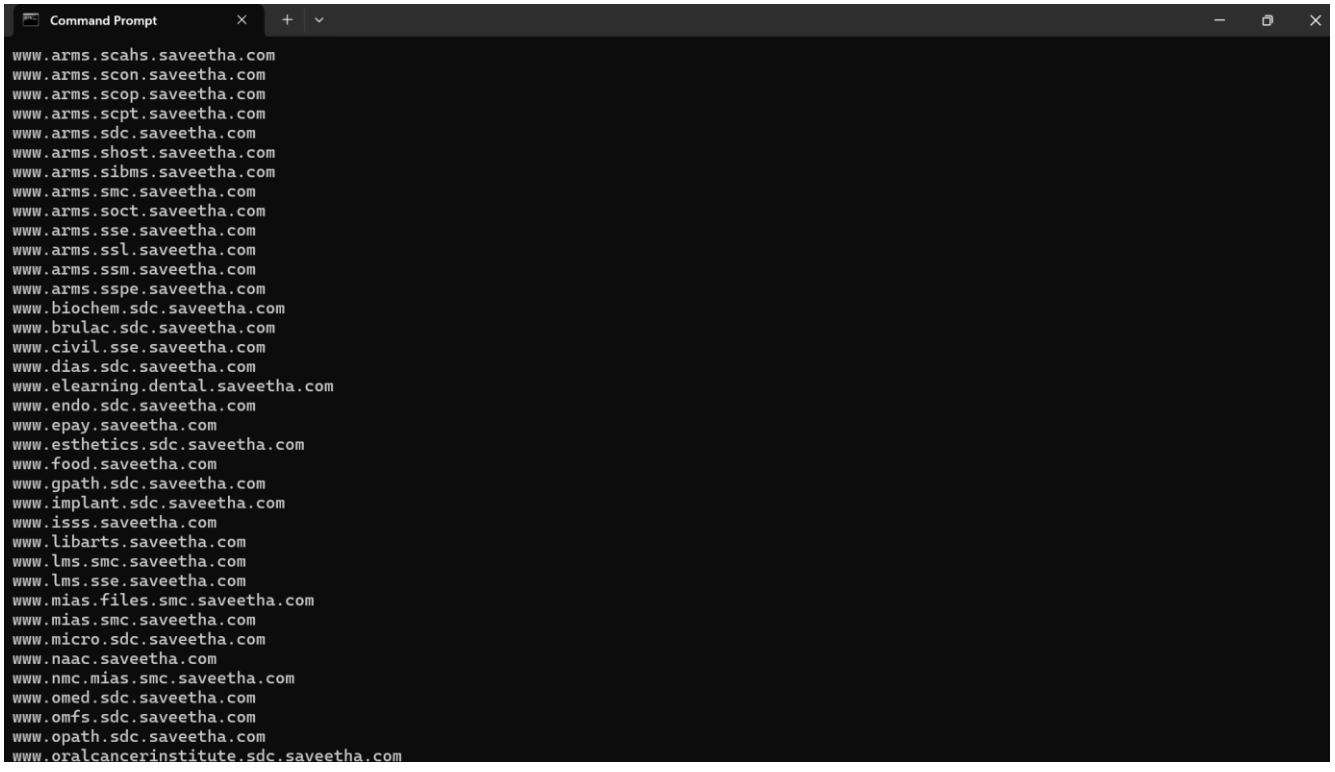
[*] Hosts found: 211
---------------------
360.saveetha.com
360c.saveetha.com
admission.saveetha.com
admission.sse.saveetha.com
```

```
[*] Hosts found: 211
---------------------
360.saveetha.com
360c.saveetha.com
admission.saveetha.com
admission.sse.saveetha.com
admissionc.sse.saveetha.com
admissions.saveetha.com
anat.sdc.saveetha.com
anat5.sdc.saveetha.com
api.vbuy.saveetha.com
arms.libarts.saveetha.com
arms.scad.saveetha.com
arms.scahs.saveetha.com
arms.scon.saveetha.com
arms.scop.saveetha.com
arms.scpt.saveetha.com
arms.sdc.saveetha.com
arms.shost.saveetha.com
arms.sibms.saveetha.com
arms.smc.saveetha.com
arms.soct.saveetha.com
arms.sse.saveetha.com
arms.ssl.saveetha.com
arms.ssm.saveetha.com
arms.sspe.saveetha.com
arms1.sse.saveetha.com
arms7.scpt.saveetha.com
armsa.ssl.saveetha.com
armsa.sspe.saveetha.com
armsc.sdc.saveetha.com
armse.scop.saveetha.com
armse.smc.saveetha.com
biochem.sdc.saveetha.com
brulac.sdc.saveetha.com
civil.sse.saveetha.com
```

```
civil.sse.saveetha.com
cse.sse.saveetha.com
dental.saveetha.com
dias.sdc.saveetha.com
diasfiles.saveetha.com
elearning.dental.saveetha.com
empowher.saveetha.com
endo.sdc.saveetha.com
endog.sdc.saveetha.com
epay.saveetha.com
epayi7.saveetha.com
esthetics.sdc.saveetha.com
files.smc.saveetha.com
food.saveetha.com
gpath.sdc.saveetha.com
gpath5.sdc.saveetha.com
hostel.saveetha.com
implant.sdc.saveetha.com
isss.saveetha.com
libarts.saveetha.com
libartsa.saveetha.com
life.saveetha.com
lms.sse.saveetha.com
mail.saveetha.com
mias.files.smc.saveetha.com
mias.smc.saveetha.com
micro.sdc.saveetha.com
micro7.sdc.saveetha.com
naac.saveetha.com
naacw.saveetha.com
nmc.mias.smc.saveetha.com
omed.sdc.saveetha.com
omfs.sdc.saveetha.com
omfsm.sdc.saveetha.com
online.saveetha.com
opath.sdc.saveetha.com
opathw.sdc.saveetha.com
```

```
omfsm.sdc.saveetha.com
online.saveetha.com
opath.sdc.saveetha.com
opathw.sdc.saveetha.com
oralcancerinstitute.sdc.saveetha.com
oralcancerinstituteg.sdc.saveetha.com
ortho.sdc.saveetha.com
orthoi.sdc.saveetha.com
pacs.saveetha.com
pacs.vhealth.saveetha.com
pedo.sdc.saveetha.com
pedo1.sdc.saveetha.com
perio.sdc.saveetha.com
periok.sdc.saveetha.com
pharm.sdc.saveetha.com
pharm7.sdc.saveetha.com
phd.sdc.saveetha.com
physio.sdc.saveetha.com
portal.saveetha.com
portala.saveetha.com
prime.sdc.saveetha.com
prostho.sdc.saveetha.com
said.sdc.saveetha.com
scad.saveetha.com
scady.saveetha.com
scahs.saveetha.com
scahs7.saveetha.com
scet.saveetha.com
scn.saveetha.com
scon.saveetha.com
scop.saveetha.com
scot.saveetha.com
scotv.saveetha.com
scp.saveetha.com
scpt.saveetha.com
scpxe.saveetha.com
sdc.saveetha.com
```

```
sdc.saveetha.com
sdcg.saveetha.com
sdg.saveetha.com
shost.saveetha.com
sibms.saveetha.com
simatsadmissions.saveetha.com
simatsentrance.saveetha.com
simatsexam.saveetha.com
simatslab.saveetha.com
simatstransport.saveetha.com
smartcard.saveetha.com
smc.saveetha.com
smcc.saveetha.com
soct.saveetha.com
sse.saveetha.com
ssl.saveetha.com
sslc.saveetha.com
ssm.saveetha.com
sspe.saveetha.com
thepupil.saveetha.com
vbuy.saveetha.com
vconnect.saveetha.com
vhealth.saveetha.com
vquest.saveetha.com
vstay.saveetha.com
vstudy.saveetha.com
vtrip.saveetha.com
webmail.saveetha.com
whitelab.sdc.saveetha.com
www.admission.sse.saveetha.com
www.anat.sdc.saveetha.com
www.arms.libarts.saveetha.com
www.arms.scad.saveetha.com
www.arms.scahs.saveetha.com
www.arms.scon.saveetha.com
www.arms.scop.saveetha.com
www.arms.scpt.saveetha.com
```



```
www.oralcancerinstitute.sdc.saveetha.com
www.ortho.sdc.saveetha.com
www.pedo.sdc.saveetha.com
www.perio.sdc.saveetha.com
www.pharm.sdc.saveetha.com
www.phd.sdc.saveetha.com
www.physio.sdc.saveetha.com
www.portal.saveetha.com
www.prostho.sdc.saveetha.com
www.said.sdc.saveetha.com
www.saveetha.com
www.scad.saveetha.com
www.scahs.saveetha.com
www.scn.saveetha.com
www.scon.saveetha.com
www.scot.saveetha.com
www.scp.saveetha.com
www.scpt.saveetha.com
www.sdc.saveetha.com
www.simatstransport.saveetha.com
www.smartcard.saveetha.com
www.smc.saveetha.com
www.sse.saveetha.com
www.ssl.saveetha.com
www.ssm.saveetha.com
www.sspe.saveetha.com
www.thepupil.saveetha.com
www.vstudy.saveetha.com
www.whitelab.sdc.saveetha.com
www1.sdc.saveetha.com
www3.physio.sdc.saveetha.com
www5.arms.sibms.saveetha.com
www9.brulac.sdc.saveetha.com
www9.gpath.sdc.saveetha.com
www9.scahs.saveetha.com
wwwa.implant.sdc.saveetha.com
wwwa.ssm.saveetha.com
```

```
www.phd.sdc.saveetha.com
www.physio.sdc.saveetha.com
www.portal.saveetha.com
www.prostho.sdc.saveetha.com
www.said.sdc.saveetha.com
www.saveetha.com
www.scad.saveetha.com
www.scahs.saveetha.com
www.scn.saveetha.com
www.scon.saveetha.com
www.scot.saveetha.com
www.scp.saveetha.com
www.scpt.saveetha.com
www.sdc.saveetha.com
www.simatstransport.saveetha.com
www.smartcard.saveetha.com
www.smc.saveetha.com
www.sse.saveetha.com
www.ssl.saveetha.com
www.ssm.saveetha.com
www.sspe.saveetha.com
www.thepupil.saveetha.com
www.vstudy.saveetha.com
www.whitelab.sdc.saveetha.com
www1.sdc.saveetha.com
www3.physio.sdc.saveetha.com
www5.arms.sibms.saveetha.com
www9.brulac.sdc.saveetha.com
www9.gpath.sdc.saveetha.com
www9.scahs.saveetha.com
wwwa.implant.sdc.saveetha.com
wwwa.ssm.saveetha.com
wwwc.scad.saveetha.com
wwwe.sse.saveetha.com
wwwu.saveetha.com

C:\Users\malli\theHarvester>
```

# RESULTS AND OBSERVATIONS

## 10. Result

TheHarvester was executed successfully enough to produce footprinting results for the target domain `saveetha.com`. Some information sources returned warnings or errors, so the results may be incomplete.

## 11. Observed Results

| Particulars | Observed result |
|---|---|
| Target domain | `saveetha.com` |
| ASNs reported | AS16509, AS53831 |
| URLs found | 2 |
| URLs displayed | `http://saveetha.com/`, `https://www.saveetha.com/` |
| IP addresses reported by the tool | 23 |
| Email addresses found | 0 |
| Hostnames found | 211 |

**Note:** The tool reported 23 IP addresses, although only 17 individual IP addresses were included in the output shared for this report. The complete IP list should be copied from the original terminal output if required.

## 12. Sample Output

The following is a summary of the observed output:

```
Target: saveetha.com

ASNs found:
AS16509
AS53831

URLs found:
http://saveetha.com/
https://www.saveetha.com/

IP addresses found: 23
Emails found: 0
Hosts found: 211
```

*This is a summarized representation of the output, not a complete terminal transcript.*

## 13. Observation and Interpretation

The experiment identified two ASNs and two URLs in the reported results. The tool also reported 23 IP addresses and 211 hostnames associated with the footprinting search.

No email addresses were discovered by the configured search during this execution. This does not prove that the organization has no email addresses; it only means that the tool did not find any through the sources and conditions used.

The results demonstrate that a domain footprinting tool can collect several categories of information in a single execution. However, the findings must be checked carefully because some entries may be incomplete, outdated, duplicated, or unrelated to the organization's current infrastructure.

# ANALYSIS, LIMITATIONS AND PRECAUTIONS

## 14. Security Analysis

Footprinting is useful for identifying information that an organization exposes publicly. Security teams can use these findings to understand their external digital footprint and improve their security practices.

**1. IP addresses:** Discovered IP addresses can help authorized security teams identify infrastructure associated with a domain. They should not be treated as proof of a vulnerability.

**2. Hostnames:** A large number of hostnames may indicate multiple websites, services, or subdomains. Organizations should maintain an inventory of their public-facing assets and review unnecessary exposures.

**3. Email addresses:** Public email addresses can be targeted by phishing and spam campaigns. Organizations should provide employee security awareness training and use appropriate email security controls.

**4. URLs:** URLs help identify publicly indexed web resources. Security teams should check that public pages do not accidentally expose confidential information.

**5. ASNs:** ASNs provide information about network routing and the autonomous systems associated with discovered infrastructure. They can help analysts understand the broader network footprint.

## 15. Limitations

The experiment had the following limitations:

- Some information sources returned errors.
- Certain sources required API credentials that were not configured.
- Windows-related Playwright or asynchronous execution warnings appeared.
- The results depended on the availability of public information sources.
- The collected information may not represent the complete infrastructure of the target domain.
- The absence of discovered email addresses does not mean that no email addresses exist.

Therefore, the findings should be considered a preliminary footprinting report rather than a complete security assessment.

## 16. Precautions

1. Perform footprinting only for academic purposes or with proper authorization.
2. Use publicly available information and follow applicable laws and institutional rules.
3. Do not attempt unauthorized access to discovered systems.
4. Do not use collected information to conduct phishing, impersonation, or other harmful activities.
5. Verify the accuracy and relevance of collected information before drawing conclusions.
6. Store screenshots and reports responsibly.

# COUNTERMEASURES AND CONCLUSION

## 17. Countermeasures

Organizations can reduce unnecessary information exposure through the following measures:

- **Public information review:** Regularly review websites and publicly indexed files to ensure that confidential information is not exposed.
- **Email security:** Use spam filtering, anti-phishing controls, multi-factor authentication (MFA), and employee awareness training.
- **Asset management:** Maintain an up-to-date inventory of public-facing domains, subdomains, IP addresses, and services.
- **Access control:** Ensure that internal or restricted resources are protected with appropriate authentication and authorization.
- **Monitoring:** Monitor public-facing assets and investigate unexpected or unauthorized exposures.
- **Secure configuration:** Remove unnecessary public services and restrict access to sensitive resources.

These measures help organizations reduce risks associated with information disclosure and misuse of publicly available data.

## 18. Learning Outcomes

After completing this experiment, I learned:

1. The meaning and importance of domain and email footprinting.
2. The purpose and basic operation of theHarvester tool.
3. How to specify a target domain and execute the footprinting command.
4. How to identify and interpret URLs, IP addresses, hostnames, email addresses, and ASNs in the output.
5. Why footprinting results must be verified and handled ethically.

## 19. Conclusion

Domain and email footprinting was performed on `saveetha.com` using theHarvester tool. The tool reported two ASNs, two URLs, 23 IP addresses, 211 hostnames, and no email addresses in the collected results. Some information sources produced errors or required API credentials, which may have limited the findings.

The experiment demonstrated how footprinting supports the reconnaissance phase of ethical hacking by identifying publicly available information about a domain. It also highlighted the importance of reviewing public information exposure and following ethical and legal guidelines during cybersecurity activities.

**Final Result:** Domain and email footprinting using theHarvester was performed, and the available results were collected and analyzed.

**RUBRICS**

| Criteria | Weightage (%) | Level 4 (Exemplary) | Level 3 (ProYicient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| **Tool Installation and ConYiguration** | 25% | Successfully installs, veriTies, and conTigures theHarvester and required environment without errors. | Installs and conTigures the tool correctly with minor issues. | Completes installation with assistance and encounters several conTiguration issues. | Unable to install, conTigure, or verify the tool. |
| **Footprinting Data Collection** | 25% | Accurately collects and organizes emails, sub-emails, hosts, IP addresses, and subdomains comprehensively. | Collects most required footprinting information with minor omissions. | Collects limited information or misses several required Tields. | Fails to obtain meaningful footprinting information. |
| **Analysis and Interpretation of Results** | 30% | Thoroughly analyzes theHarvester output and accurately interprets the signiTicance of discovered information. | Correctly interprets most results with minor errors. | Provides basic interpretation with limited technical understanding. | Unable to interpret the collected footprinting information. |
| **Evidence and Documentation** | 20% | Provides complete screenshots, counts, lists, and organized documentation with technically accurate presentation. | Provides adequate evidence and documentation with minor omissions. | Documentation is partially complete or poorly organized. | Evidence is missing, incomplete, or difTicult to verify. |